NO NONDEGENERATE SIMPLICES OF I_{M!}U ABOVE DEGREE M

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: self-contained given Lemma lemma-simplicial-set-n-skel-sub and the adjunction i_{m!} -| sk_m; the author's alternative route via Lemma lemma-left-adjoint-exists (not registered) is only mentioned as an aside and is not needed.

Editorial note. The label lemma-imshriek-sets names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Generated subobject U'. Same unit, Lemma lemma-simplicial-set-n-skel-sub. The subobject V' used in the argument is the one of Lemma lemma-simplicial-set-n-skel-sub.

Source reference. Original reference 018L: sources/stacks-simplicial/pinned-original-source.tex, source lines 3152–3152. Prerequisite pointer invoked at proof line(s) 3240; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 017T: sources/stacks-simplicial/pinned-original-source.tex, source lines 2074–2074. Prerequisite pointer invoked at proof line(s) 3245; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *If U is an m-truncated simplicial set and $n > m$ then all n-simplices of $i_{m!}U$ are degenerate.*

Proof. This can be seen from the construction of $i_{m!}U$ in Lemma 018L, but we can also argue directly as follows. Write $V = i_{m!}U$. Let $V' \subset V$ be the simplicial subset with $V'_i = V_i$ for $i \leq m$ and all i simplices degenerate for $i > m$, see Lemma 017T. By the adjunction formula, since $\mathrm{sk}_m V' = U$, there is an inverse to the injection $V' \to V$. Hence $V' = V$. □

Date: stacks-project:simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.